QWeatherRiskF

Data fixes report: what was fixed and why

Item	Detail
Starting data	merged_district_year_v1b.csv (ICRISAT / IMD-derived, 13 AP districts, 2015-2019)
Final dataset	merged_cleaned_v1f.csv (65 x 151)
Crop-long file	issue1/crop_long_v1.csv (1,430 rows: district-year-crop)
Audit trail	v1f_change_log.csv (every cell changed by the fix plan: 104 cells with old value, new value, reason)

1. Summary

- Eight items were examined: five changed values, one removed columns, and two kept their values but were excluded from modelling. Every change is evidence-based, logged and flagged. Against the original v1b, 667 cells differ for documented reasons and 0 for any other reason.
- **Fixed with proof:** sugarcane units (x10, matches official totals within 0.1%), cotton 2019 (production was ~19% of the official figure; area was correct), minor pulses 2018-19 (zeros meant "not reported").
- **Estimated, flagged:** cotton 2019 in 8 small districts, from the official combined total, split by area x prior yield (back-test total error about 10-15%; area-only about 27-30%). The combined total is preserved exactly.
- **Excluded, not repaired:** the oilseeds aggregate (it includes oil palm and other crops) and the sorghum aggregate (redundant); the minor pulses group. They stay in the long file with modelling_role = exclude_*.
- **Missing cells fall from 757 (v1b) to 247 (v1f)**: 169 are the structural rabi_prev 2015 gap and 78 are minor pulses 2018-19 (excluded from modelling). Every other cell is filled, official or flagged as an estimate.

2. Dataset lineage

Version	What changed	Missing cells
v1b	Original merged file	757
filledv2	Srikakulam rainfall filled (data owner update); rain_valid_area_fraction 0.75 to 1.0	641
v1c	+ sugarcane production and yield x10 for 2015-17 (78 cells)	641
v1d	Team cleaning: blanks filled with 0, barley and 6 area-only series dropped, columns renamed to snake_case	169
v1f	+ cotton 2019 corrected (5 official, 8 estimated), minor pulses 2018-19 set to missing (104 cells)	247

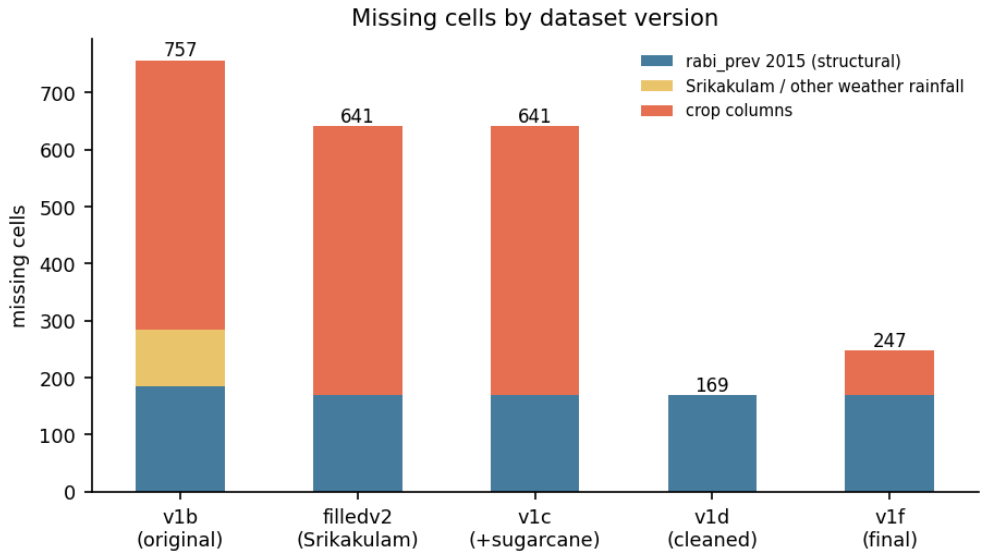


Figure 1. Missing cells by version. The rise at v1f is the 78 minor pulses cells set to missing on purpose.

3. Fix register: what, why, evidence

#	Issue	Evidence	Action	Cells	Status
1	Srikakulam rainfall missing (all years)	Not random: a coverage rule suppressed it. New values are consistent with neighbours (mean 1,221 mm vs 1,140-1,157 mm; rainy days correlate 0.97 with Vizianagaram) and with the rabi_prev(t) = rabi_next(t-1) identity	Filled by data owner (filledv2)	121	Done; fill method to be documented
2	Sugarcane production and yield jump about 10x in 2018	Dataset production = 0.1 x official in 2015-17 (ratio 0.1000, 0.1000, 0.0999); x10 gives 9,353 vs official 9,353 thousand t; area already matched	x10 for 2015-17	78	Done (v1c); no jump remains
3	Agricultural blanks (NaN) in 2016-17	97.7% of blanks fall in 2016-17; same crop-districts are 0 in other years; 153 of 179 blank records have every other year exactly 0, the rest tiny areas (max 0.5 thousand ha)	Blank set to 0; flagged (flag_nan_filled_with_zero)	365	Done (v1d); yield 0 where area 0 is undefined
4	Crops with no signal	Barley all zero/blank; six area-only series have no production or yield	Dropped (9 columns)	-	Done (v1d)
5	Cotton 2019 production and yield	Area matches official in every year (ratio 1.00); production only 19% of official (80 vs 426 thousand t; official 25.08 lakh bales); ratio 0.18-0.22 in five main districts; not a constant factor	5 districts replaced with official; 8 estimated by allocation; yield recomputed	25	Done; 8 values are estimates
6	Minor pulses 2018-19 = 0	12 of 13 districts exactly 0 after 600-740 thousand ha in 2015-17; crops still grown in rabi 2018-19 and 2019-20 per AP crop coverage reports	Set to missing; exclude from modelling	78	Done
7	Oilseeds total vs 8 members	Mismatch in 35 of 65 rows (7 districts, every year); excess area tracks official oil palm area (Spearman 0.74, n = 40) and implied yield 15,000-25,000 kg/ha matches oil palm bunches	Kept, flagged, excluded from modelling	0	Done
8	Sorghum aggregate	Equals kharif + rabi sorghum in all rows; redundant	Kept, flagged, excluded from modelling	0	Done

Cell accounting against the original v1b (all changed cells assigned to a documented fix):

Reason	Cells changed
Srikakulam rainfall filled (data-owner update)	121
Sugarcane x10 (2015-17 production, yield)	78
Agricultural blanks converted to 0	365
Cotton 2019 (replaced / estimated)	25
Minor pulses 2018-19 set to missing	78
Other	0
Total	667

In the table above, Srikakulam counts every weather cell that differs from v1b (rain variables across windows, plus rain_valid_area_fraction); "Other" is 0 where no unexplained change exists.

4. Plots: before and after

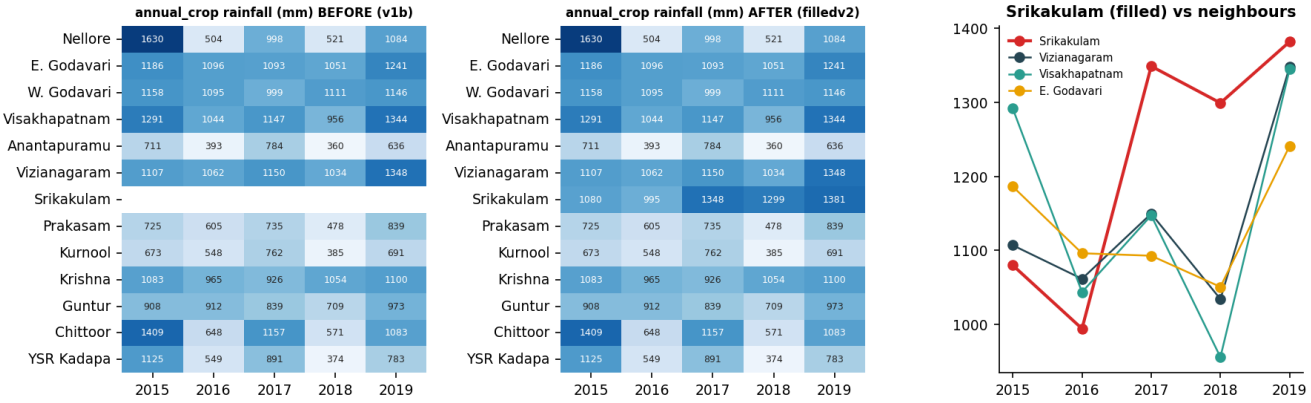


Figure 2. Fix 1: annual rainfall by district before and after the Srikakulam update, and Srikakulam against its neighbours.

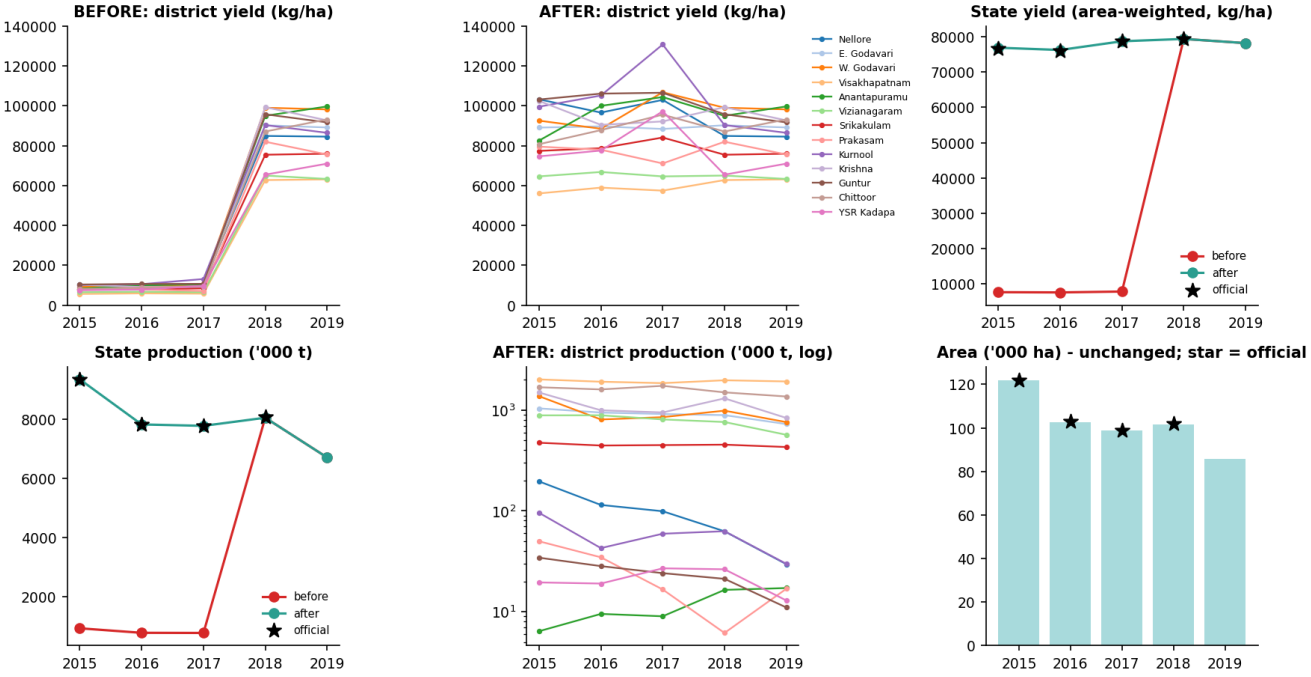


Figure 3. Fix 2: sugarcane before and after the x10 correction. Stars are official figures.

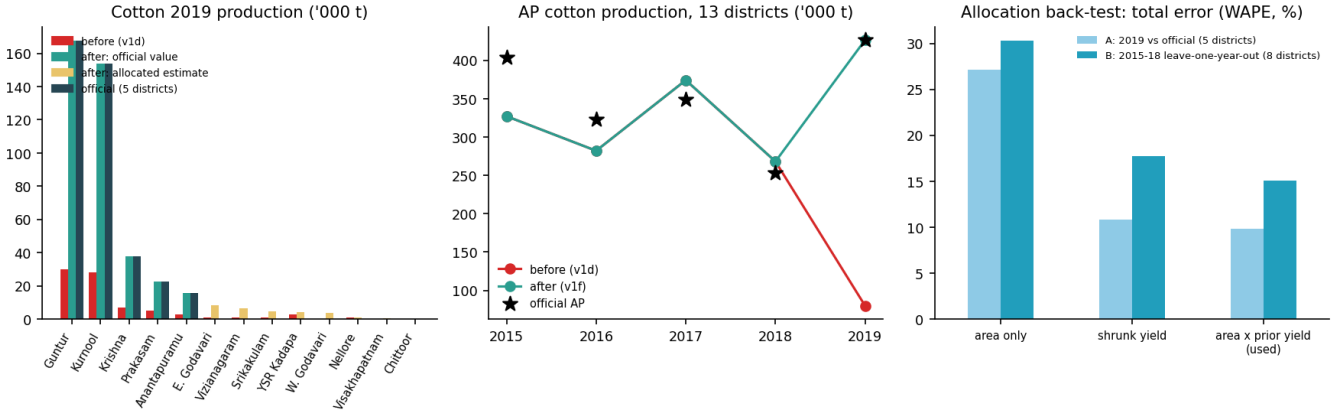


Figure 4. Fix 5: cotton 2019 production by district (left), state production against official (centre), and the allocation back-test (right).

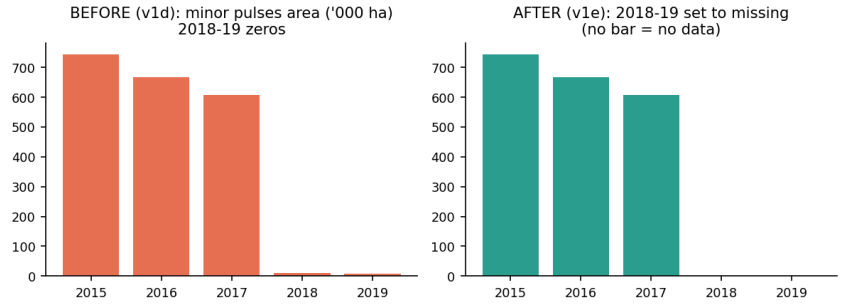


Figure 5. Fix 6: minor pulses area before and after (2018-19 set to missing).

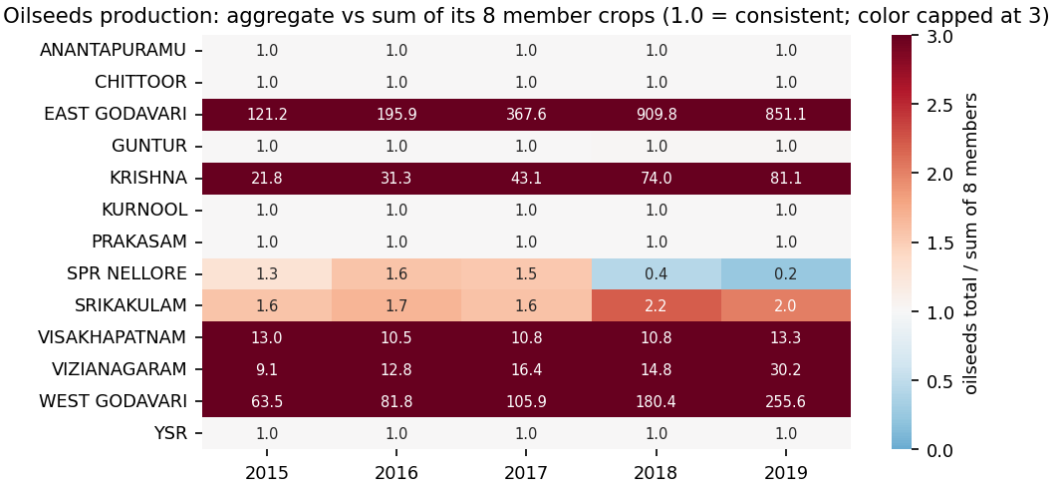


Figure 6. Fix 7: oilseeds total divided by the sum of its eight members (1.0 = consistent). The aggregate is excluded from modelling.

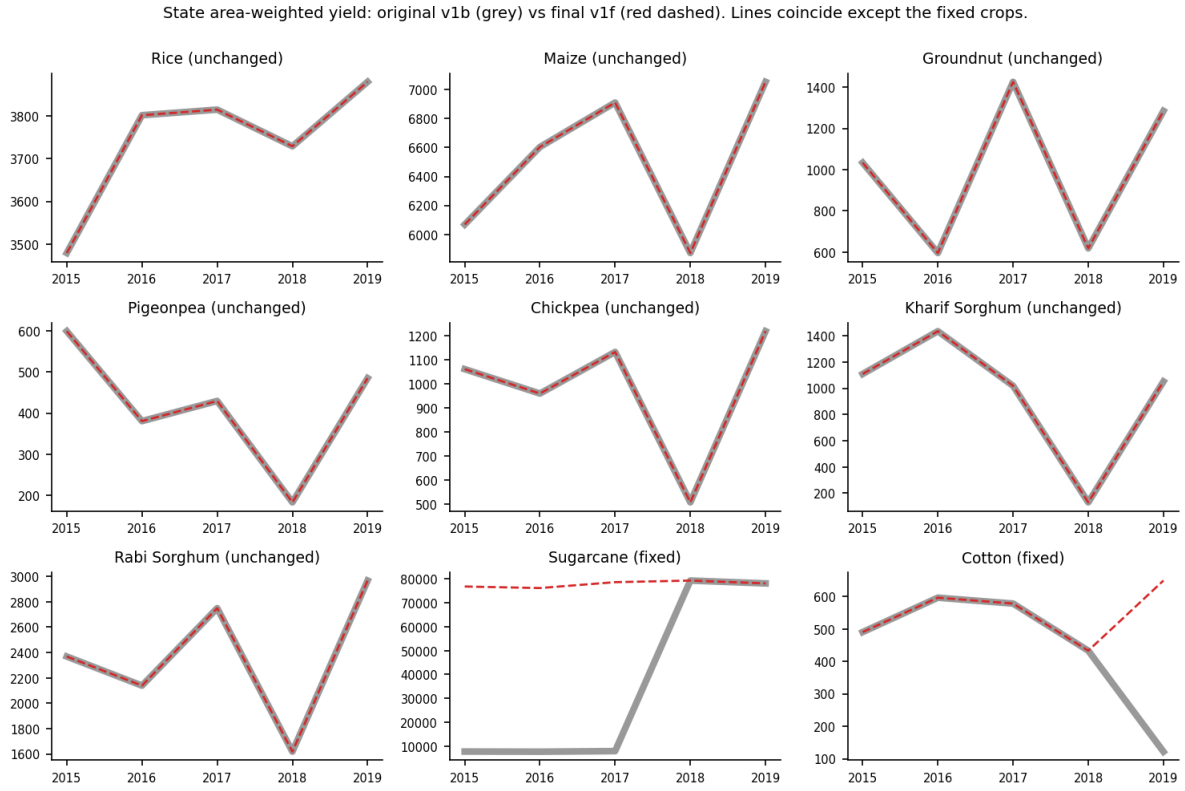


Figure 7. No other crop changed: state area-weighted yield for the main crops, original v1b against final v1f.

5. Cotton 2019 allocation (how the eight missing districts were filled)

Only the combined official production of the eight small districts is published (1.71 lakh bales = 29.07 thousand tonnes). It was split in proportion to **area (2019) x prior yield**, where prior yield is the district's area-weighted reported yield for 2015-2018. This is a standard pro-rata disaggregation: the combined total is kept exactly (29.06 against 29.07, rounding), and district differences come from their own history instead of assuming all districts have the same yield. Two back-tests were run: (A) pretend only the five main districts' combined 2019 total is known and compare the split with their official values, and (B) for 2015-18, split the eight districts' totals using priors from the other years. The chosen method had a total error of 9.8% (A) and 15.1% (B); allocation by area alone had 27.1% and 30.3%.

District	Area 2019 (000 ha)	Estimated production (000 t)	Estimated yield (kg/ha)
Srikakulam	6	4.76	793
Visakhapatnam	1	0.34	340
E. Godavari	13	8.18	629
W. Godavari	5	3.87	774
Nellore	2	1.12	560
YSR Kadapa	15	4.34	289
Chittoor	0	0.00	0
Vizianagaram	12	6.45	538

These are estimates, flagged in the long file (flag_estimated_by_allocation). Per-district error is about 15% on average and larger for the smallest districts. Chittoor has zero recorded area, so its production is 0. The estimates should be replaced as soon as the official AP district table is obtained.

6. What to use for modelling and what remains open

- **Use:** the 19 crops with use_in_modelling = True in crop_long_v1.csv. Keep the flags; filter flag_estimated_by_allocation and flag_low_area_lt_1000ha rows in sensitivity analyses.
- **Do not use:** oilseeds, sorghum (aggregate) and minor pulses.
- **Open:** (a) tie-out of other crops in 2018-19 against official series, since the problems found all begin in 2018; (b) official AP district cotton table for 2019; (c) yield stored as 0 where area is 0 (treat as undefined); (d) rabi_prev is structurally missing for 2015 and redundant with rabi_next of the previous year; (e) documentation of weather window dates, units and the Srikakulam fill method.
- **Caveats:** the replacement and allocation inputs come from secondary compilations of AP statistics (ANGRAU, ICAR-CICR); please confirm them against the primary Directorate of Economics and Statistics tables before external citation.